

ADVANCED

Stage 6

Trigonometry (Adv), T3 Trig Functions and Graphs (Adv)

Trig Applications (Y12)

Teacher: SHS Maths

Exam Equivalent Time: 82.5 minutes (based on allocation of 1.5 minutes per mark)

T3 Trig Functions and Graphs



■ Trig Graphs
■ Trig Applications

*SmarterMaths analytics based on the average contribution to new syllabus Advanced Maths exams since 2020.

HISTORICAL CONTRIBUTION

- T3 Trig Functions and Graphs* has contributed a meaningful 5.8% per new syllabus Adv exam since being introduced in 2020.
- This topic has been split into two sub-topics for analysis purposes: *1-Trig Graphs (1.8%), and 2-Trig Applications (4.0%)*.
- This analysis looks at the largest sub-topic, *Trig Applications*.

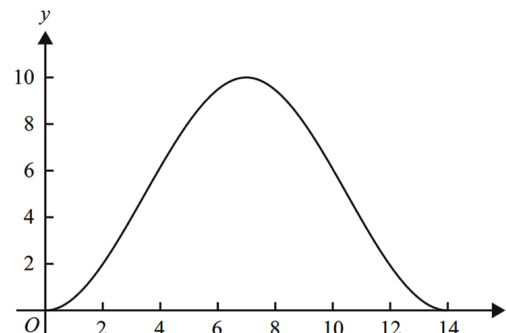
HSC ANALYSIS - What to expect and common pitfalls

- Trig Applications (4.0%)* did not appear in the 2023 HSC exam after being examined with substantial allocations of 6-7 marks each year in the period 2020-22 (note the 2021 question was a large cross topic example incorporating integration which will be covered in a later topic).
- This is a very interesting topic area in our view. While only examined 5 times since 2009, a number of these questions are in the pantheon of great Advanced HSC exam questions during this period.
- Early indications from new syllabus mark allocations are that students can expect these type of questions appearing more regularly.
- With a high difficulty level and substantial mark allocations almost a guarantee when asked, students should revise this topic area diligently.

Questions

1. Trigonometry, 2ADV T3 SM-Bank 5 MC

The UV index, y , for a summer day in Newcastle East is illustrated in the graph below, where t is the number of hours after 6 am.



The graph is most likely to be the graph of

- $y = 5 + 5\cos\left(\frac{\pi t}{7}\right)$
- $y = 5 - 5\cos\left(\frac{\pi t}{7}\right)$
- $y = 5 + 5\cos\left(\frac{\pi t}{14}\right)$
- $y = 5 - 5\cos\left(\frac{\pi t}{14}\right)$

2. Trigonometry, 2ADV T3 SM-Bank 13

On any given day, the depth of water in a river is modelled by the function

$$h(t) = 14 + 8\sin\left(\frac{\pi t}{12}\right), \quad 0 \leq t \leq 24$$

where h is the depth of water, in metres, and t is the time, in hours, after 6 am.

- Find the minimum depth of the water in the river. (1 mark)
- Find the values of t for which $h(t) = 10$. (2 marks)

3. Trigonometry, 2ADV T3 SM-Bank 15

The graphs of $y = \cos(x)$ and $y = a\sin(x)$, where a is a real constant, have a point of intersection at $x = \frac{\pi}{3}$.

- Find the value of a . (2 marks)
- Find the x -coordinate of the other point of intersection of the two graphs, given $0 \leq x \leq 2\pi$ (1 mark)

4. Trigonometry, 2ADV T3 2018 HSC 15a

The length of daylight, $L(t)$, is defined as the number of hours from sunrise to sunset, and can be modelled by the equation

$$L(t) = 12 + 2\cos\left(\frac{2\pi t}{366}\right),$$

where t is the number of days after 21 December 2015, for $0 \leq t \leq 366$.

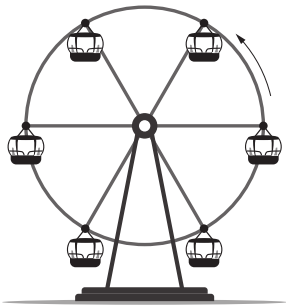
- Find the length of daylight on 21 December 2015. (1 mark)
- What is the shortest length of daylight? (1 mark)
- What are the two values of t for which the length of daylight is 11? (2 marks)

5. Trigonometry, 2ADV T3 2024 HSC 28

Anna is sitting in a carriage of a Ferris wheel which is revolving. The height, $A(t)$, in metres above the ground of the top of her carriage is given by

$$A(t) = c - k\cos\left(\frac{\pi t}{24}\right),$$

where t is the time in seconds after Anna's carriage first reaches the bottom of its revolution and c and k are constants.



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The top of each carriage reaches a greatest height of 39 metres and a smallest height of 3 metres.

- Find the value of c and k . (2 marks)
- How many seconds does it take for one complete revolution of the Ferris wheel? (1 mark)
- Billie is in another carriage. The height, $B(t)$, in metres above the ground of the top of her carriage is given by

$$B(t) = c - k\cos\left(\frac{\pi}{24}(t - 6)\right),$$

where c and k are as found in part (a).

During each revolution, there are two occasions when Anna's and Billie's carriages are at the same heights. At what two heights does this occur? Give your answer correct to 2 decimal places. (4 marks)

6. Trigonometry, 2ADV T3 SM-Bank 10

The population of wombats in a particular location varies according to the rule

$$n(t) = 1200 + 400\cos\left(\frac{\pi t}{3}\right),$$

where n is the number of wombats and t is the number of months after 1 March 2018.

- Find the period and amplitude of the function n . (2 marks)
- Find the maximum and minimum populations of wombats in this location. (2 marks)
- Find $n(10)$. (1 mark)
- Over the 12 months from 1 March 2018, find the fraction of time when the population of wombats in this location was less than $n(10)$. (2 marks)

7. Trigonometry, 2ADV T3 2013 HSC 13a

The population of a herd of wild horses is given by

$$P(t) = 400 + 50\cos\left(\frac{\pi}{6}t\right)$$

where t is time in months.

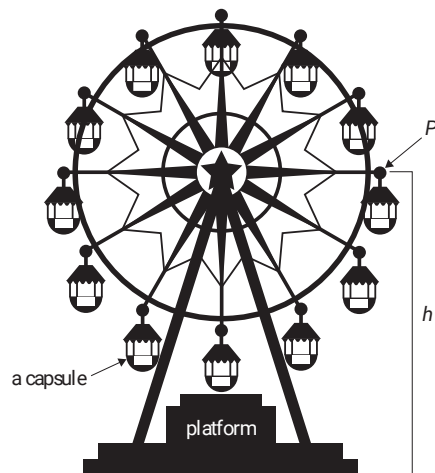
- Find all times during the first 12 months when the population equals 375 horses. (2 marks)
- Sketch the graph of $P(t)$ for $0 \leq t \leq 12$. (2 marks)

8. Trigonometry, 2ADV T3 SM-Bank 16

Sammy visits a giant Ferris wheel. Sammy enters a capsule on the Ferris wheel from a platform above the ground. The Ferris wheel is rotating anticlockwise. The capsule is attached to the Ferris wheel at point P .

The height of P above the ground, h , is modelled by $h(t) = 65 - 55\cos\left(\frac{\pi t}{15}\right)$, where t is the time in minutes after Sammy enters the capsule and h is measured in metres.

Sammy exits the capsule after one complete rotation of the Ferris wheel.



- State the minimum and maximum heights of P above the ground. (1 mark)
- For how much time is Sammy in the capsule? (1 mark)
- Find the rate of change of h with respect to t and, hence, state the value of t at which the rate of change of h is at its maximum. (2 marks)

9. Trigonometry, 2ADV T3 2022 HSC 23

The depth of water in a bay rises and falls with the tide. On a particular day the depth of the water, d metres, can be modelled by the equation

$$d = 1.3 - 0.6\cos\left(\frac{4\pi}{25}t\right),$$

where t is the time in hours since low tide.

- Find the depth of water at low tide and at high tide. (2 marks)
- What is the time interval, in hours, between two successive low tides? (1 mark)
- For how long between successive low tides will the depth of water be at least 1 metre? (3 marks)

10. Trigonometry, 2ADV T3 2011 SPEC1 8

Find the coordinates of the points of intersection of the graph of the relation

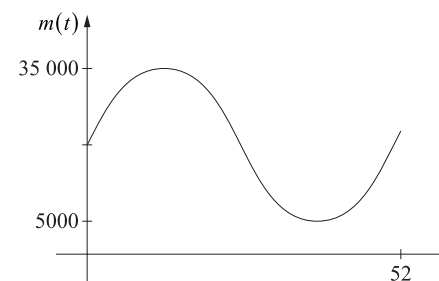
$$y = \operatorname{cosec}^2\left(\frac{\pi x}{6}\right) \text{ with the line } y = \frac{4}{3}, \text{ for } 0 < x < 12. \quad (3 \text{ marks})$$

11. Trigonometry, 2ADV T3 2020 HSC 31

The population of mice on an isolated island can be modelled by the function.

$$m(t) = a\sin\left(\frac{\pi}{26}t\right) + b,$$

where t is the time in weeks and $0 \leq t \leq 52$. The population of mice reaches a maximum of 35 000 when $t = 13$ and a minimum of 5000 when $t = 39$. The graph of $m(t)$ is shown.



- What are the values of a and b ? (2 marks)
- On the same island, the population of cats can be modelled by the function

$$c(t) = -80\cos\left(\frac{\pi}{26}(t - 10)\right) + 120$$

Consider the graph of $m(t)$ and the graph of $c(t)$.

Find the values of t , $0 \leq t \leq 52$, for which both populations are increasing. (3 marks)

- Find the rate of change of the mice population when the cat population reaches a maximum. (2 marks)

12. Trigonometry, 2ADV T3 2009 HSC 7b

Between 5 am and 5 pm on 3 March 2009, the height, h , of the tide in a harbour was given by

$$h = 1 + 0.7\sin\left(\frac{\pi}{6}t\right) \text{ for } 0 \leq t \leq 12$$

where h is in metres and t is in hours, with $t = 0$ at 5 am.

- What is the period of the function h ? (1 mark)
- What was the value of h at low tide, and at what time did low tide occur? (2 marks)
- A ship is able to enter the harbour only if the height of the tide is at least 1.35 m.
Find all times between 5 am and 5 pm on 3 March 2009 during which the ship was able to enter the harbour. (3 marks)

Worked Solutions

1. Trigonometry, 2ADV T3 SM-Bank 5 MC

Centre line (median): $y = 5$

Amplitude = 5

$$\text{Period: } 14 = \frac{2\pi}{n}$$
$$n = \frac{\pi}{7}$$

$$\therefore \text{Graph: } y = 5 - 5\cos\left(\frac{\pi t}{7}\right)$$

$$\Rightarrow B$$

2. Trigonometry, 2ADV T3 SM-Bank 13

i. $h_{\min}$ occurs when $\sin\left(\frac{\pi t}{12}\right) = -1$

$$\therefore h_{\min} = 14 - 8$$
$$= 6 \text{ m}$$

ii. $14 + 8\sin\left(\frac{\pi}{12}t\right) = 10$

$$\sin\left(\frac{\pi}{12}t\right) = -\frac{1}{2}$$

Solve in general:

$$\frac{\pi}{12}t = \frac{7\pi}{6} + 2\pi n \quad \text{or} \quad \frac{\pi}{12}t = \frac{11\pi}{6} + 2\pi n,$$
$$t = 14 + 24n \qquad t = 22 + 24n$$

Substitute integer values for n ,

$$\therefore t = 14 \text{ or } 22, \quad (0 \leq t \leq 24)$$

MARKER'S COMMENT: Students who used calculus to find the minimum were less successful.

Worked Solutions

3. Trigonometry, 2ADV T3 SM-Bank 15

i. Intersection occurs when $x = \frac{\pi}{3}$,

$$a\sin\left(\frac{\pi}{3}\right) = \cos\left(\frac{\pi}{3}\right)$$

$$\tan\left(\frac{\pi}{3}\right) = \frac{1}{a}$$

$$\sqrt{3} = \frac{1}{a}$$

$$\therefore a = \frac{1}{\sqrt{3}}$$

ii. $\tan(x) = \sqrt{3}$

$$x = \frac{\pi}{3}, \frac{4\pi}{3}, 2\pi + \frac{\pi}{3}, \dots$$

$$\therefore x = \frac{4\pi}{3} \quad (0 \leq x \leq 2\pi)$$

4. Trigonometry, 2ADV T3 2018 HSC 15a

i. $L(t) = 12 + 2\cos\left(\frac{2\pi t}{366}\right)$

On 21 Dec 2015 $\Rightarrow t = 0$

$\therefore L(0) = 12 + 2\cos 0$

$= 14 \text{ hours}$

ii. Shortest length of daylight occurs when

$\cos\left(\frac{2\pi t}{366}\right) = -1$

$\therefore \text{Shortest length} = 12 + 2(-1)$

$= 10 \text{ hours}$

iii. Find t such that $L(t) = 11$:

$11 = 12 + 2\cos\left(\frac{2\pi t}{366}\right)$

$\cos\left(\frac{2\pi t}{366}\right) = -\frac{1}{2}$

$\frac{2\pi t}{366} = \frac{2\pi}{3} \quad \text{or} \quad \frac{2\pi t}{366} = \frac{4\pi}{3}$

$t = \frac{366}{3} \quad t = \frac{366 \times 2}{3}$

$= 122 \quad = 244$

$\therefore t = 122 \text{ or } 244$

♦ Mean mark 43%.

5. Trigonometry, 2ADV T3 2024 HSC 28

a. $c = \text{centre of motion} = \frac{39 + 3}{2} = 21$

$k = \text{amplitude} = \frac{39 - 3}{2} = 18$

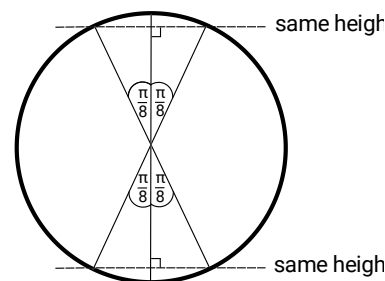
b. $A(t) = 21 - 18\cos\left(\frac{\pi t}{24}\right) \Rightarrow n = \frac{\pi}{24}$

$T = \frac{2\pi}{n} = 2\pi \times \frac{24}{\pi} = 48 \text{ seconds}$

c. Billie's carriage is 6 seconds behind Anna's.

$\text{Angle between the 2 carriages} = \frac{\pi \times 6}{24} = \frac{\pi}{4}$

By inspection:



Heights are the same:

$h_1 = 21 - 18\cos\left(\frac{\pi}{8}\right) = 4.370\dots = 4.37 \text{ m (2 d.p.)}$

$h_2 = 21 - 18\cos\left(\frac{7\pi}{8}\right) = 37.629\dots = 37.63 \text{ m (2 d.p.)}$

6. Trigonometry, 2ADV T3 SM-Bank 10

i. $\text{Period} = \frac{2\pi}{n} = \frac{2\pi}{\frac{\pi}{3}} = 6 \text{ months}$

$\text{Amplitude} = 400$

ii. $\text{Max: } 1200 + 400 = 1600 \text{ wombats}$

$\text{Min: } 1200 - 400 = 800 \text{ wombats}$

iii. $n(10) = 1200 + 400\cos\left(\frac{10\pi}{3}\right)$

$$= 1200 + 400\cos\left(\frac{2\pi}{3}\right)$$

$$= 1200 - 400 \times \frac{1}{2}$$

$$= 1000 \text{ wombats}$$

iv. Find t when $n(t) = 1000$

$$1000 = 1200 + 400\cos\left(\frac{\pi t}{3}\right)$$

$$\cos\left(\frac{\pi t}{3}\right) = -\frac{1}{2}$$

$$\frac{\pi t}{3} = \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{8\pi}{3}, \frac{10\pi}{3}, \dots$$

$$t = 2, 4, 8, 10$$

Since $n(0) = 1600$,

$\Rightarrow n(t)$ drops below 1000 between $t = 2$ and $t = 4$,
and between $t = 8$ and $t = 10$.

$$\therefore \text{Fraction} = \frac{2+2}{12}$$

$$= \frac{1}{3} \text{ year}$$

7. Trigonometry, 2ADV T3 2013 HSC 13a

i. $P(t) = 400 + 50\cos\left(\frac{\pi}{6}t\right)$

Need to find t when $P(t) = 375$

$$375 = 400 + 50\cos\left(\frac{\pi}{6}t\right)$$

$$50\cos\left(\frac{\pi}{6}t\right) = -25$$

$$\cos\left(\frac{\pi}{6}t\right) = -\frac{1}{2}$$

Since $\cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$, and $\cos$ is

negative in 2nd/3rd quadrants:

$$\Rightarrow \frac{\pi}{6}t = \left(\pi - \frac{\pi}{3}\right), \left(\pi + \frac{\pi}{3}\right), \left(3\pi - \frac{\pi}{3}\right)$$

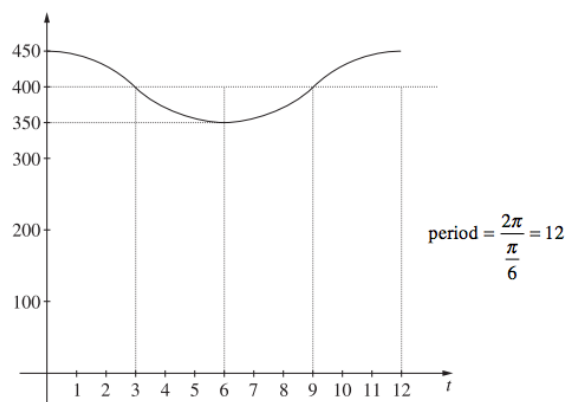
$$= \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{8\pi}{3}, \dots$$

$$\therefore t = 4, 8, 16, \dots$$

$\therefore$ In the 1st 12 months, $P(t) = 375$ when

$t = 4$ months and 8 months.

ii.



♦ Mean mark 39%

8. Trigonometry, 2ADV T3 SM-Bank 16

$$\text{i. } h_{\min} = 65 - 55 \quad h_{\max} = 65 + 55 \\ = 10 \text{ m} \quad = 120 \text{ m}$$

$$\text{ii. Period} = \frac{2\pi}{\frac{\pi}{15}} = 30 \text{ min}$$

$$\text{iii. } h(t) = 65 - 55 \cos\left(\frac{\pi t}{15}\right) \\ h(t) = \frac{\pi}{15} \times 55 \sin\left(\frac{\pi}{15}t\right) \\ = \frac{11\pi}{3} \sin\left(\frac{\pi}{15}t\right)$$

$$\text{Since } \sin\left(\frac{\pi}{15}t\right)_{\max} = \sin\left(\frac{\pi}{2}\right),$$

$$\therefore h(t)_{\max} \text{ occurs when}$$

$$\frac{\pi t}{15} = \frac{\pi}{2}$$

$$\therefore t = \frac{\pi}{2} \times \frac{15}{\pi} \\ = \frac{15}{2} \text{ minutes } (0 \leq t \leq 30)$$

9. Trigonometry, 2ADV T3 2022 HSC 23

$$\text{a. Since } -1 \leq \cos\left(\frac{4\pi}{25}t\right) \leq 1:$$

$$\text{Low Tide} = 1.3 - 0.6(1) = 0.7 \text{ m}$$

$$\text{High Tide} = 1.3 - 0.6(-1) = 1.9 \text{ m}$$

$$\text{b. Time between two low tides} = \text{Period of equation } (n)$$

$$\frac{2\pi}{n} = \frac{4\pi}{25}$$

$$\frac{n}{2\pi} = \frac{25}{4\pi}$$

$$n = \frac{25}{2} \text{ hours}$$

$$\text{c. Find } t \text{ when } d = 1:$$

$$1.3 - 0.6 \cos\left(\frac{4\pi}{25}t\right) = 1$$

$$-0.6 \cos\left(\frac{4\pi}{25}t\right) = -0.3$$

$$\cos\left(\frac{4\pi}{25}t\right) = \frac{1}{2}$$

$$\frac{4\pi}{25}t = \frac{\pi}{3}, \frac{5\pi}{3}$$

$$t = \frac{25}{12}, \frac{125}{12}$$

$$\therefore \text{Time between low tides where water depth} \geq 1 \text{ m}$$

$$= \frac{125}{12} - \frac{25}{12}$$

$$= \frac{100}{12}$$

$$= \frac{25}{3} \text{ hours}$$

♦ Mean mark part (b)
47%.

♦ Mean mark part (c)
46%.

10. Trigonometry, 2ADV T3 2011 SPEC1 8

Intersection occurs when:

$$\operatorname{cosec}^2\left(\frac{\pi x}{6}\right) = \frac{4}{3}$$

$$\operatorname{cosec}\left(\frac{\pi x}{6}\right) = \pm \frac{2}{\sqrt{3}}$$

$$\sin\left(\frac{\pi x}{6}\right) = \pm \frac{\sqrt{3}}{2}$$

$$\text{Given: } 0 < x < 12 \Rightarrow 0 < \frac{\pi x}{6} < 2\pi$$

$$\frac{\pi x}{6} = \frac{\pi}{3}, \pi - \frac{\pi}{3}, \pi + \frac{\pi}{3}, 2\pi - \frac{\pi}{3}$$

$$= \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$$

$$x = 2, 4, 8, 10$$

$$\Rightarrow y = \frac{4}{3} \text{ for each}$$

$$\therefore \text{Intersection at: } \left(2, \frac{4}{3}\right), \left(4, \frac{4}{3}\right), \left(8, \frac{4}{3}\right), \left(10, \frac{4}{3}\right)$$

11. Trigonometry, 2ADV T3 2020 HSC 31

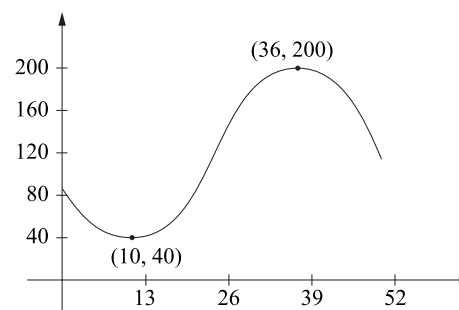
$$\begin{aligned} \text{a. } b &= \frac{35\,000 + 5000}{2} \\ &= 20\,000 \end{aligned}$$

$$\begin{aligned} a &= \text{amplitude of sin graph} \\ &= 35\,000 - 20\,000 \\ &= 15\,000 \end{aligned}$$

b. By inspection of the $m(t)$ graph

$$m'(t) > 0 \text{ when } 0 \leq t < 13 \text{ and } 39 < t \leq 52$$

Sketch $c(t)$:



Minimum ($\cos 0$) when $t = 10$

Maximum ($\cos \pi$) when $t = 36$

$$\therefore c'(t) > 0 \text{ when } 10 < t < 36$$

$\therefore$ Both populations are increasing when $10 < t < 13$

c. $c(t)$ maximum when $t = 36$

$$m(t) = 15\,000 \sin\left(\frac{\pi}{26}t\right) + 20\,000$$

$$m'(t) = \frac{15\,000\pi}{26} \cos\left(\frac{\pi}{26}t\right)$$

$$\begin{aligned} m'(36) &= \frac{15\,000\pi}{26} \cdot \cos\left(\frac{36\pi}{26}\right) \\ &= -642.7 \end{aligned}$$

$\therefore$ Mice population is decreasing at 643 mice per week.

♦♦ Mean mark part (b) 30%.

♦♦♦ Mean mark part (c) 27%.

12. Trigonometry, 2ADV T3 2009 HSC 7b

i. $h = 1 + 0.7\sin\left(\frac{\pi}{6}t\right)$ for $0 \leq t \leq 12$

$$\begin{aligned} T &= \frac{2\pi}{n} \text{ where } n = \frac{\pi}{6} \\ &= 2\pi \times \frac{6}{\pi} \\ &= 12 \text{ hours} \end{aligned}$$

$\therefore$ The period of h is 12 hours.

ii. Find h at low tide

$\Rightarrow h$ will be a minimum when

$$\sin\left(\frac{\pi}{6}t\right) = -1$$

$$\begin{aligned} \therefore h_{\min} &= 1 + 0.7(-1) \\ &= 0.3 \text{ metres} \end{aligned}$$

IMPORTANT: Using $\sin x = -1$ for a minimum here is very effective and time efficient. This property of trig functions is *often very useful* in harder questions.

Since $\sin x = -1$ when $x = \frac{3\pi}{2}$

$$\begin{aligned} \frac{\pi}{6}t &= \frac{3\pi}{2} \\ t &= \frac{3\pi}{2} \times \frac{6}{\pi} \\ &= 9 \text{ hours} \end{aligned}$$

$\therefore$ Low tide occurs at 2pm (5 am + 9 hours)

iii. Find t when $h \geq 1.35$

$$1 + 0.7\sin\left(\frac{\pi}{6}t\right) \geq 1.35$$

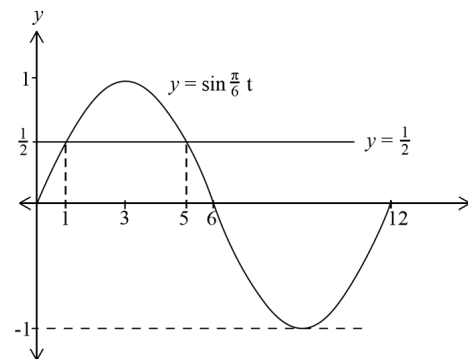
$$0.7\sin\left(\frac{\pi}{6}t\right) \geq 0.35$$

$$\sin\left(\frac{\pi}{6}t\right) \geq \frac{1}{2}$$

$$\sin\left(\frac{\pi}{6}t\right) = \frac{1}{2} \text{ when}$$

$$\frac{\pi}{6}t = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \text{ etc } \dots$$

$$t = 1, 5 \quad (0 \leq t \leq 12)$$



From the graph,

$$\sin\left(\frac{\pi}{6}t\right) \geq \frac{1}{2} \text{ when } 1 \leq t \leq 5$$

$\therefore$ Ship can enter the harbour between 6 am and 10 am.